

Data Intake Processing and Verification Report

Background Information

- **Original Dataset Name:** BlueFlux: Modeled Daily CO₂ and CH₄ Wetland Fluxes, Southern Florida, 2000-2024
- **GHG Center Dataset Title:** BlueFlux CO₂ and CH₄ Fluxes for Southern Florida Wetlands
- **Dataset Provider:** NASA
- **Date Obtained:** June 2025
- **Location Obtained From:**
 - <https://doi.org/10.3334/ORNLDAAAC/2404>
- **Data Location in GHG Center:** blueflux-ghgflux-daygrid-v1
- **Data POC(s):** Dr. Ayia Lindquist, Dr. Cheryl Doughty
- **Dataset File Type(s):** NetCDF
- **Projection (if different from WGS84):** N/A

Data Transfer Confirmation

An SHA-256 checksum is used to detect high-level errors within data transmissions. Results from individual checksum file comparisons of pre-transfer and post-transfer shows all files were transferred successfully and no individual files had any transfer issues.

Data Intake Process

- https://us-ghg-center.github.io/ghgc-docs/data_workflow/blueflux-ghgflux-daygrid-v1_Data_Flow.html

Methane Dataset Statistics

NetCDF (original data) and COGs (transformed Cloud Optimized GeoTIFF) files for 5 randomly chosen dates were read to generate statistics to validate transformed data against the original data files.

Date	Type	Min	Mean	Max	Std
2000-03-01	NetCDF(raw)	-38.72	28.78	163.62	18.67
	COG (transformed)	-38.72	28.78	163.62	18.67
2005-08-01	NetCDF(raw)	-41.28	44.22	165.15	21.47
	COG (transformed)	-41.28	44.22	165.15	21.47

2010-12-01	NetCDF(raw)	-38.38	31.32	140.34	18.5
	COG (transformed)	-38.38	31.32	140.34	18.5
2015-05-01	NetCDF(raw)	-29.49	31.06	152.22	19.82
	COG (transformed)	-29.49	31.06	152.22	19.82
2024-01-01	NetCDF(raw)	-38.45	39.39	147.45	19.72
	COG (transformed)	-38.45	39.39	147.43	19.72

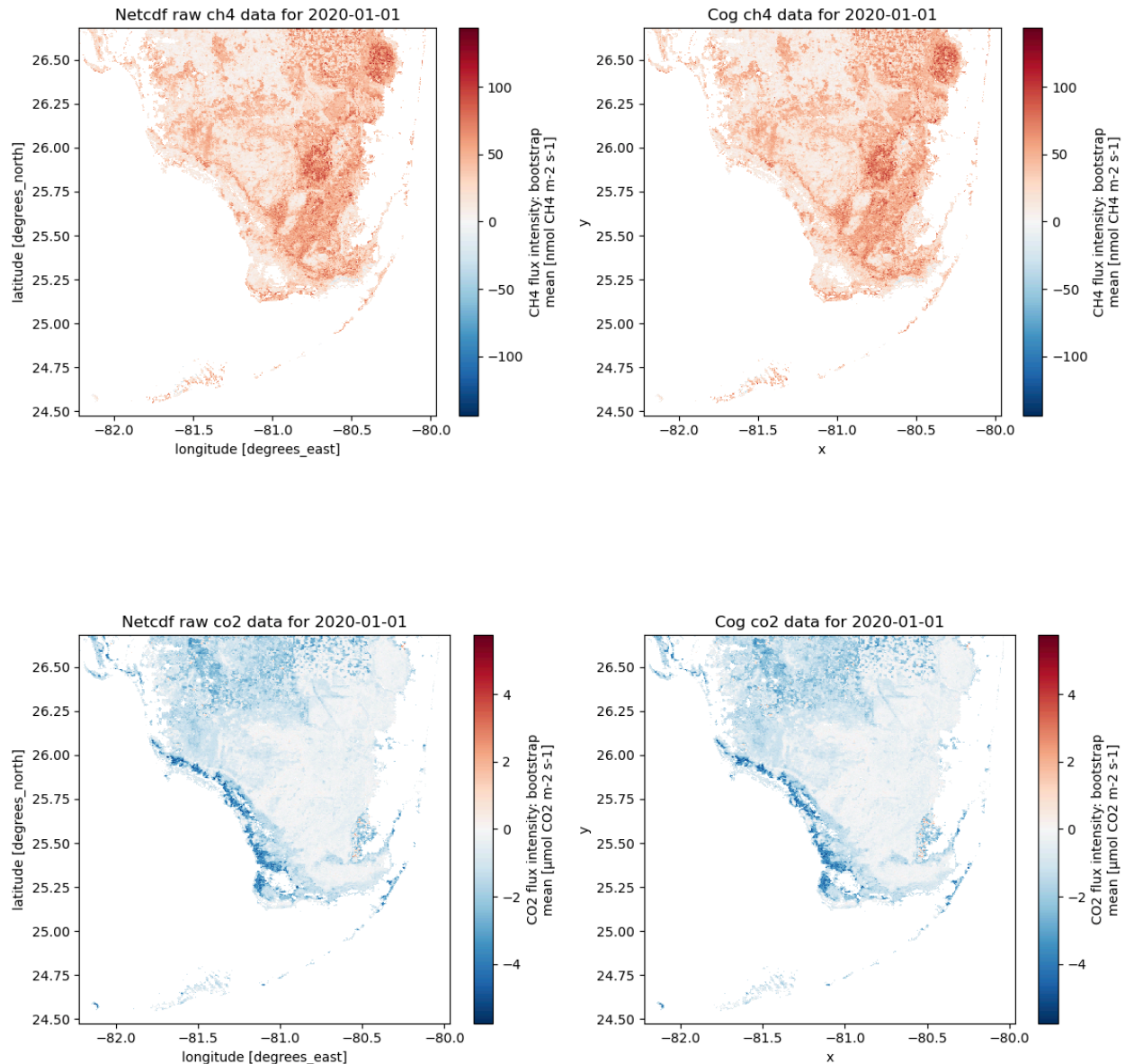
Carbon Dioxide Dataset Statistics

NetCDF (original data) and COGs (transformed Cloud Optimized GeoTIFF) files for 5 randomly chosen dates were read to generate statistics to validate transformed data against the original data files.

Date	Type	Min	Mean	Max	Std
2001-04-01	NetCDF(raw)	-5.53	-0.89	3.81	0.86
	COG (transformed)	-5.53	-0.89	3.81	0.86
2007-09-01	NetCDF(raw)	-5.91	-1.11	4.15	1.01
	COG (transformed)	-5.91	-1.11	4.15	1.01
2013-11-01	NetCDF(raw)	-5.91	-0.96	3.78	0.85
	COG (transformed)	-5.91	-0.96	3.78	0.85
2019-10-01	NetCDF(raw)	-5.78	-1.06	4.87	0.91
	COG (transformed)	-5.78	-1.06	4.87	0.91
2022-07-01	NetCDF(raw)	-5.88	-0.93	3.26	1.01
	COG (transformed)	-5.88	-0.93	3.26	1.01

Visual Inspection

Plots below for CO₂ and CH₄ data on **2020-01-01** have been mapped to show a visual comparison between the original files (NetCDF) and transformed files (COG).



The mean and other statistical measures of the COGs generated using the transformation script matched those of the original NetCDF files. Additionally, visual comparisons of the CO₂ and CH₄ flux data between the COGs and the NetCDF files showed no noticeable differences. Therefore, we can conclude that the data transformation did not introduce any errors.

Summary

- We are confident that the transformation and display of data in the US GHG Center accurately represents the source data
- [Link to the transformation code](#)
- [Link](#) to the dataset Tutorial
- [Link](#) to the US GHG Center data catalog page

Report Completed on: 7/15/2025

MSFC POC for questions: [Jeanne le Roux](#), [Siddharth Chaudhary](#)